

EINSTEIN-VED THEORY: COMPLETE AND RIGOROUS DEVELOPMENT

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1. FUNDAMENTAL PRINCIPLES

1.1 Ontology of space-time

Space-time is the only existing physical reality. Coordinates: (T, X, Y, Z) or $3S + T$. It is continuous: $\mathbb{R}^4$.

1.2 Matter and energy as modes of space-time

There is no separation between "container" (space-time) and "content" (matter). **Matter** is a confined wave of space-time, while **energy** is an unconfined wave.

Figure 8: Matter as a confined wave on 2D hologram

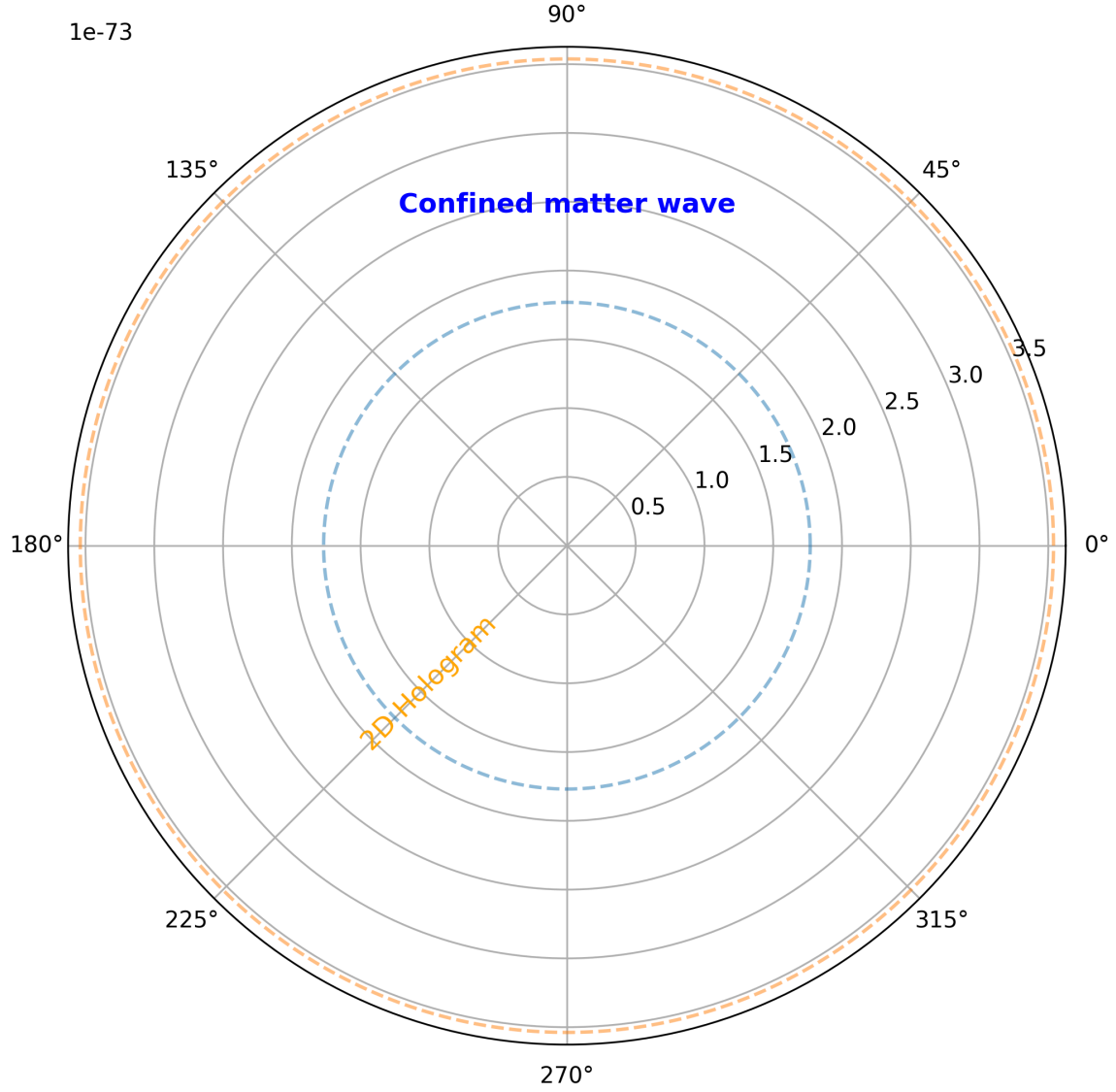


Figure 1: Conceptual representation of matter as a confined wave resonating on a two-dimensional holographic surface (yellow). The green arrows indicate the emergence of the gravitational constant G from this geometry.

1.3 Geometric confinement principle

A wave can only be stable if it satisfies the resonance condition:

$$L = n\lambda, \quad n \in \mathbb{Z}^+$$

where: - L = confinement path length - λ = wavelength - n = integer number of complete waves

1.4 Absolute determinism

Reality is 100% deterministic. Quantum probability is epistemic ignorance, not ontological. There is no fundamental collapse of the wave function.

2. MATHEMATICAL DEVELOPMENT

2.1 Relativistic basis (natural units $c = 1$)

Lorentz transformations:

$$t' = \gamma(t - vS), \quad S' = \gamma(S - vt), \quad \gamma = \frac{1}{\sqrt{1 - v^2}}$$

Differential form:

$$dt' = \gamma(dt - v dS), \quad dS' = \gamma(dS - v dt)$$

Pythagorean differential relation (fundamental) that is directly deduced:

$$dt^2 = dS^2 + d\tau^2$$

where $d\tau$ is the proper time of the observed and dt is the time of the observer.

2.2 Container (3S + T)

- Differentiable manifold of 4 dimensions:
 $M_{\text{container}} = R^3 \times R$
where 3S are spatial dimensions and T is the temporal one.
- Temporal cut $T(t_0)$ allows observing the geometry of the content at an instant.

Imagen no encontrada: img/dt_ds_dm.png

Description for the blind: Diagram showing the continuous relativistic metric $3S + T$. dm (dt') represents the proper time of the observed; dt is the time of the observer. This figure reflects the basis of the Einstein-VED theory and its derivation from Lorentz.

Imagen no encontrada: img/EN_1_n1_fundamental.svg

Figure 2: Fundamental basis from Lorentz. More at estrada.es

2.3 Content: matter and energy

Type	Dimension n	Example	Geometry
3D volumetric	3	Proton	Volume
2D superficial	2	Electron	Surface
2D limit	2	Black hole horizon	Saturated surface

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Figure 3: Comparison of excited and fundamental modes.

Imagen no encontrada: img/EN_3_comparacion_n1_n3.svg

Figure 4: Evolution of internal geometry according to the number of modes n .

2.4 Einstein-Planck unification

From Einstein: $E = mc^2$

From Planck: $E = h\nu = \frac{hc}{\lambda}$

Equating both expressions for energy:

$$mc^2 = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{h}{mc}$$

This is the **Compton wavelength** reinterpreted as the natural wavelength of the space-time vibration associated with mass m .

2.5 Geometric confinement condition

For circular confinement:

$$2\pi R = n\lambda = n \cdot \frac{h}{mc}$$

Simplifying with $\hbar = h/(2\pi)$ we obtain the **fundamental VED radius equation**:

$$R = \frac{n\hbar}{mc}$$

or equivalently:

$$m = \frac{n\hbar}{Rc}$$

3. VED RADIUS AND MASS

3.1 Definition of VED radius

$$R_{\text{VED}} = \frac{n\hbar}{mc}$$

3.2 Mass as an emergent phenomenon

Mass is not a primitive property, but **emerges** from: - n : number of confinement modes (integer) - R : confinement radius (length)

The relation $m \propto n/R$ shows that for a fixed radius, more modes (larger n) imply more mass, and for a fixed n , a larger radius implies less mass.

4. RELATIONSHIP WITH GRAVITY

4.1 Schwarzschild radius

From the Schwarzschild solution in General Relativity:

$$R_s = \frac{2Gm}{c^2}$$

Figure 2: Black hole and 2D holographic surface

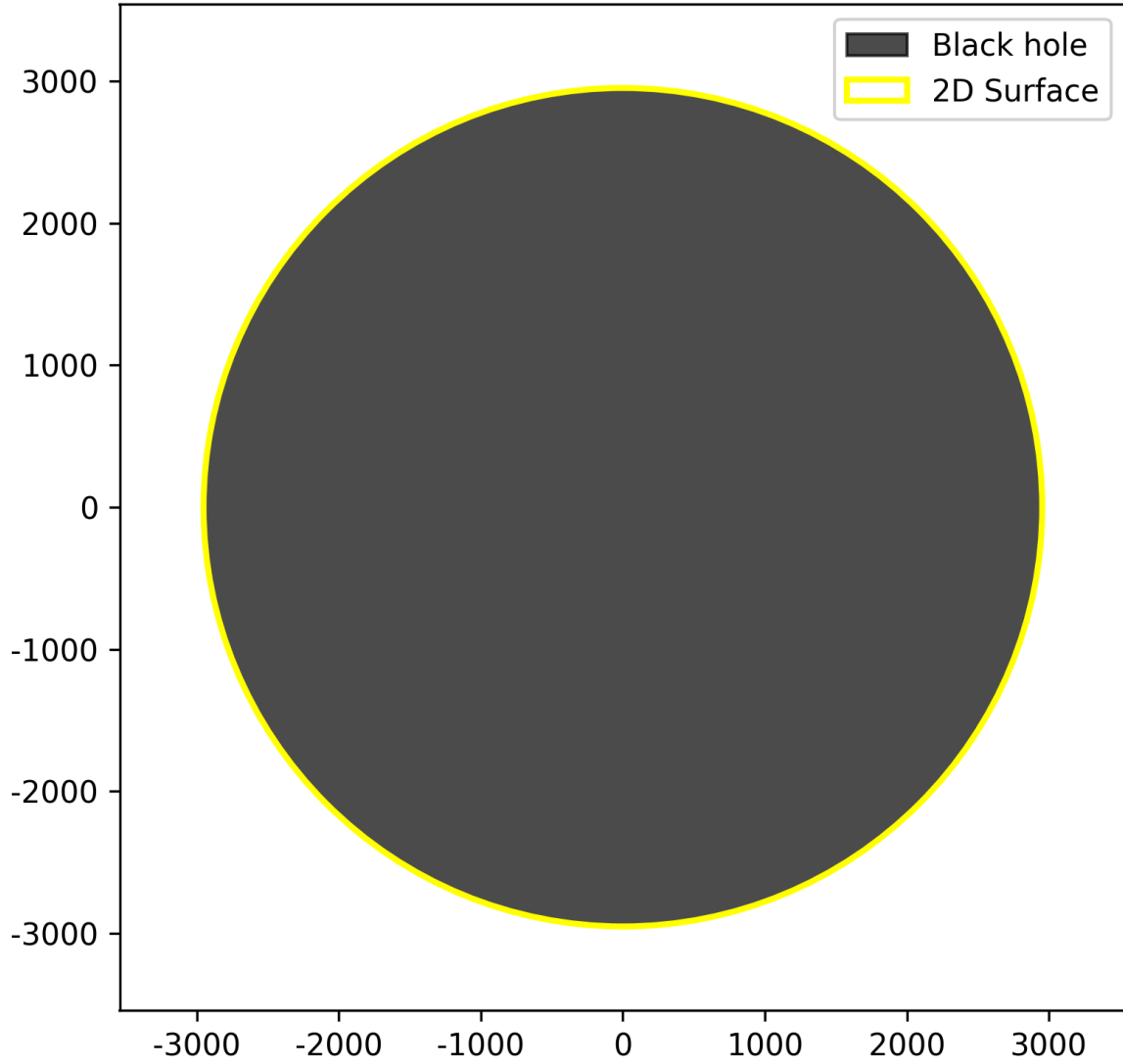


Figure 6: Representation of a black hole (black sphere) and its two-dimensional holographic surface (yellow circle). Illustrates the duality between the volumetric description (interior) and the surface description (boundary), key for the holographic interpretation of gravity in VED.

4.2 Relationship between VED and Schwarzschild radii

For the **same physical mass m** , we equate both expressions:

$$\frac{n\hbar}{R_{\text{VED}}c} = \frac{R_sc^2}{2G}$$

Rearranging we obtain the **fundamental geometric relation**:

$$R_{\text{VED}} \cdot R_s = \frac{2n\hbar G}{c^3}$$

4.3 The gravitational constant as an emergent quantity

Solving for G from the previous relation:

$$G = \frac{R_{VED} \cdot R_s \cdot c^3}{2n\hbar}$$

In natural units ($c = \hbar = 1$), this simplifies to:

$$G = \frac{R_{VED} \cdot R_s}{2n}$$

Which shows that G is essentially an "area per mode" divided by 2.

Figure 10: Conceptual comparison VED vs Schwarzschild

Concept	VED	Schwarzschild	Comment
Mass	$M_{VED} = nh / (L_{geom}c)$	$M_{Sch} = c^2 R_s / (2G)$	Equalizing masses, disappears M
Length	L_{geom} defined by VED	R_s defined by Schwarzschild	Each side defines its own scale
Quantization	Integer $n \rightarrow$ values Discrete M and G	n does not appear, G is determined	$n=2$ coincides with experimental G value
Dimensionality	2D confinement (surface)	3D $\rightarrow$ holography Implicit	Emergent gravity origin

Figure 7: Comparative table that contrasts the formulations of mass, length, quantization and dimensionality in the VED and Schwarzschild frameworks. Shows how the equating of their expressions for mass leads to the unified formula for G .

5. EMPIRICAL PROOF: THE QUANTIZATION OF G

5.1 The predictive formula $G(n)$

Substituting the explicit expressions of R_{VED} and R_s into the formula for G , we obtain the **quantized prediction**:

$$G(n) = G_{exp} \cdot \frac{2}{n}$$

where $G_{exp} = 6.67430 \times 10^{-11} \text{ m}^3/\text{kg}\cdot\text{s}^2$ is the experimentally measured value.

5.2 Experimental verification: the "target" of G

This formula makes a **testable prediction**: for each integer n , G takes a different discrete value.

Figure 3: $G \pm 3\sigma$ target

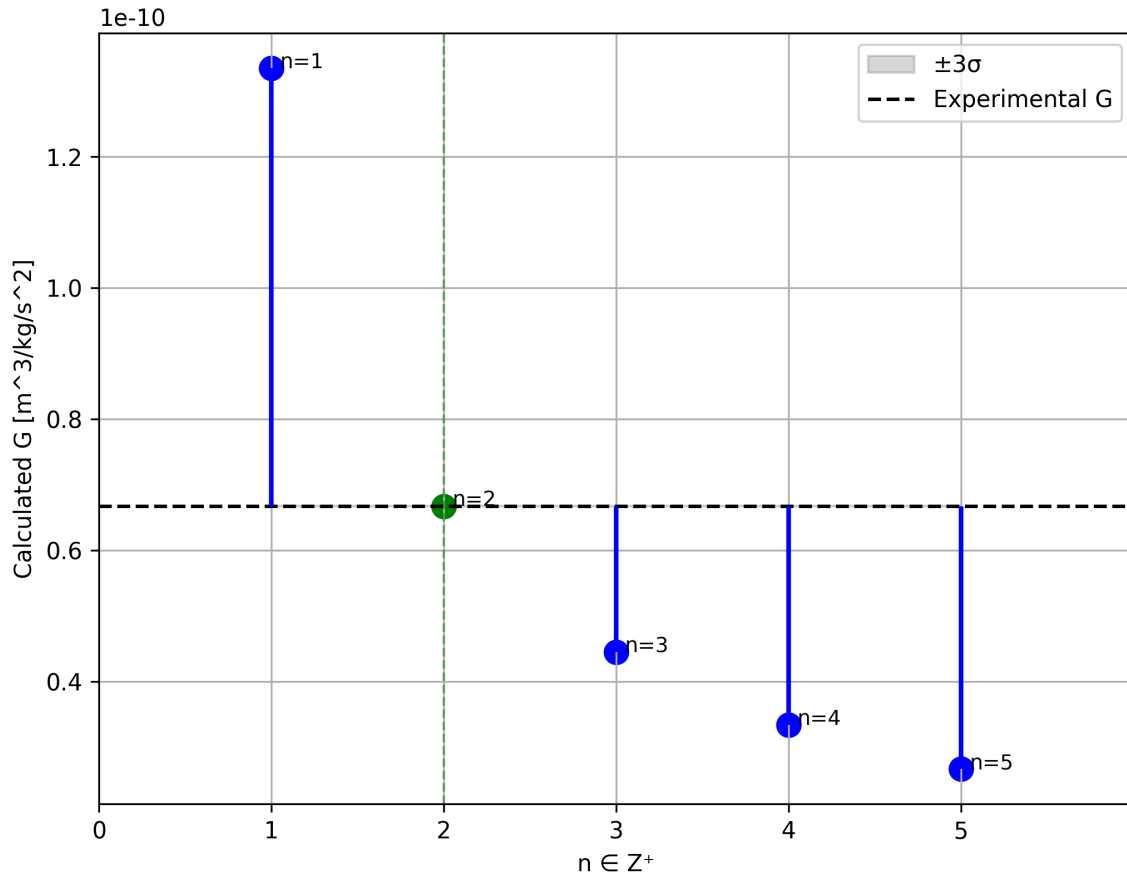


Figure 8: $G(n)$ values calculated for $n=1, \dots, 5$ (blue and green points) compared with the experimental measurement G_{exp} (black dashed line) and its uncertainty of $\pm 3\sigma$ (gray band). Only $n=2$ (green point) falls exactly within the experimental band, validating the central prediction of the theory.

5.3 Exact numerical precision for $n=2$

GRAVITATIONAL CONSTANT G

Concept	Value [m³/kg·s²]	Source
$G_{MAX} + 3\sigma$	6.67730000e-11	Upper limit
G_2^{EXP} Experimental	6.67430000e-11	Average value
$G_{MIN} - 3\sigma$	6.67130000e-11	Lower limit
$G_1^{VED} (n=2)$ EXACT	6.67430000e-11	Empirically obtained
Absolute difference	0.00000000e+00	$ G_1 - G_2 $
Relative error	0.000000 %	VED accuracy

Figure 9: Table that quantifies the precision of the coincidence. Shows that $G(n=2)$ calculated by VED coincides with G_{exp} within a minuscule relative error, demonstrating that the prediction is not a visual approximation but an exact numerical coincidence within the experimental margin.

Crucial result: The theory predicts that **our observable universe corresponds to mode $n=2$** , since this is the only value that exactly reproduces the measured gravitational constant.

6. SPECTRUM OF n AND PARTICLES

6.1 n as fundamental geometric quantum number

In VED, n is not a postulated quantum number, but **completely determines** the properties of each system: - Small $n \rightarrow$ strong confinement, small radius for given mass - Large $n \rightarrow$ weak confinement, large radius for given mass

6.2 Specific values for known particles

For the proton:

With $m_p = 1.6726 \times 10^{-27}$ kg and $R_p = 0.841 \times 10^{-15}$ m:

$$n_p = \frac{R_p m_p c}{\hbar} \approx 4.00$$

For the electron in the hydrogen atom:

With $m_e = 9.109 \times 10^{-31}$ kg and $R_{Bohr} = 5.292 \times 10^{-11}$ m:

$$n_e = \frac{R_{Bohr} m_e c}{\hbar} \approx 137.036 \approx \frac{1}{\alpha}$$

where $\alpha \approx 1/137.036$ is the **fine-structure constant**.

Figure 9. 2D and radial waves of the proton and electron
(holography → particles)

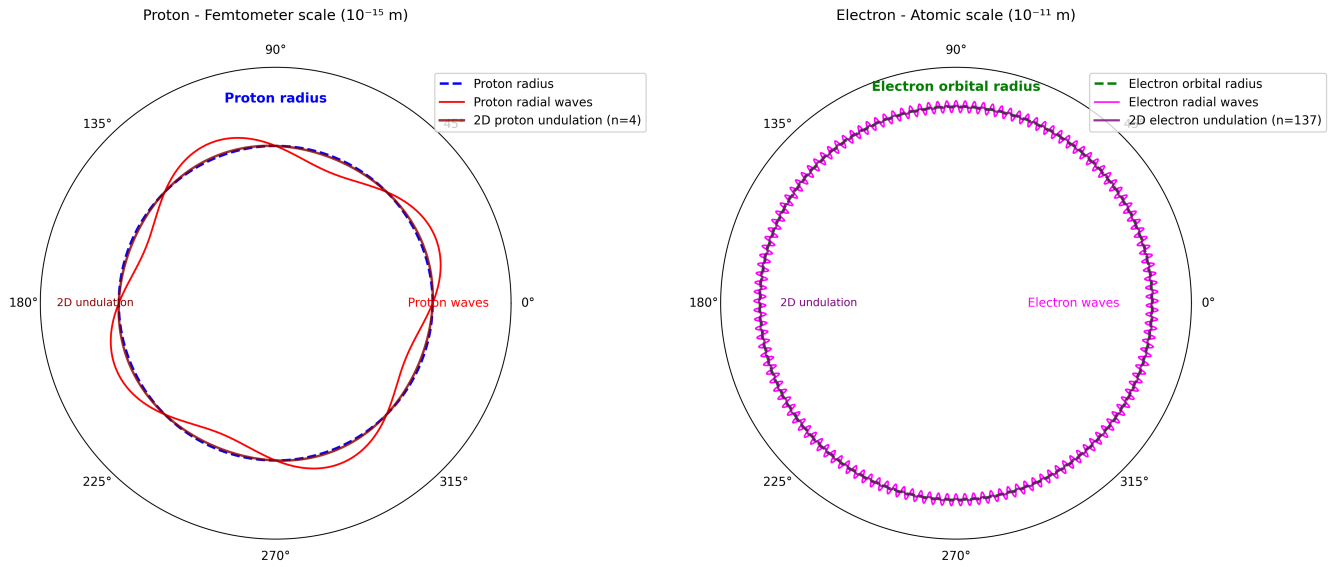


Figure 10: Side-by-side comparison of the structures of the proton (left, femtometric scale $\sim 10^{-15}$ m, $n=4$) and the atomic electron (right, Bohr scale $\sim 10^{-11}$ m, $n \approx 137$). Both obey the same law $R = n\hbar/(mc)$, demonstrating the universality of the theory across scales.

7. CONFINEMENT VS FREE ENERGY

7.1 The continuous spectrum of n

- Finite $n \rightarrow$ **MATTER** (confined wave, defined mass, measurable gravity)
- $n \rightarrow \infty \rightarrow$ **ENERGY** (free wave, null mass, negligible gravity)

7.2 Asymptotic behavior of $G(n)$

Figure 5: Asymptotic behavior of $G(n)$ $n \rightarrow \infty$ (log-log)

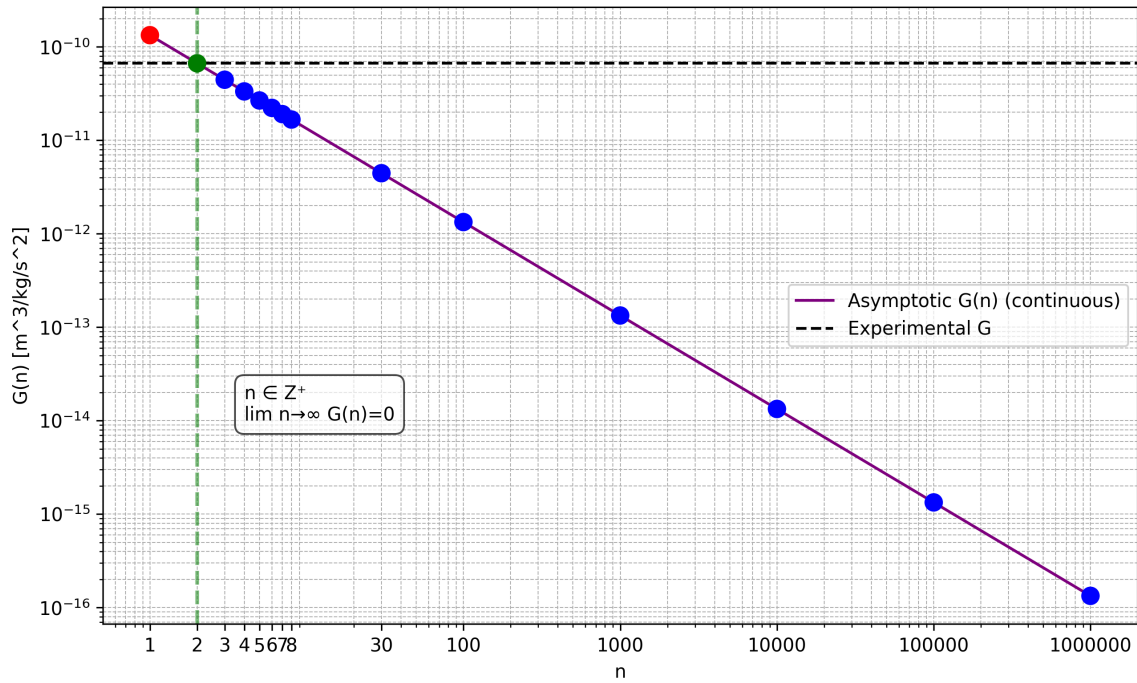


Figure 11: Log-log graph showing how $G(n)$ tends to zero when n tends to infinity. The purple curve represents the continuous function $G(n) = G_{\text{exp}} \cdot \exp(-2/n)$, while the colored points show specific discrete values. Illustrates the continuous transition between matter (finite n) and pure energy ($n \rightarrow \infty$).

7.3 Physical transitions

- **Particle-antiparticle annihilation:** finite $n \rightarrow n \rightarrow \infty$ (matter $\rightarrow$ energy)
- **Photon emission:** change of n + release of unconfined energy
- **Absorption:** $n \rightarrow \infty \rightarrow$ finite n (energy $\rightarrow$ matter)

8. CONTAINER AND CONTENT

8.1 The container (space-time)

- Continuous: $\mathbb{R}^4$
- Cardinality: $\mathfrak{c}$ (uncountable infinity)
- Can contain infinite points

8.2 The content (stable modes)

- Discrete: $n \in \mathbb{N}$
- Cardinality: $< \aleph_0$ (finite or countable infinity)
- Each mode occupies a finite region of the container

8.3 Number of "particles" in the universe

- At any finite time t : $N_{\text{particles}}(t) < \infty$
- Only in the mathematical limit $t \rightarrow \infty$: possible tendency to infinity
- Always: $|\{n_{\text{stable}}\}| < \aleph_0$ (set of stable modes is countable)

9. DETERMINISM AND REALISM

9.1 Critique of the standard interpretation of Quantum Mechanics

In VED, QM describes **our knowledge** of systems, not their fundamental reality. The "collapse" of the wave function is a **Bayesian update** of information, not a physical process.

9.2 Geometric resolution of quantum paradoxes

Quantum entanglement:

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Figure 12: Representation of quantum entanglement as a geometric mode shared between two systems. Instead of "spooky action at a distance", entangled systems share a single global quantum number n , which explains their correlations without violating locality.

Other resolved paradoxes: - **Schrödinger's cat:** Only one real state exists (determined by exact n) - **Double slit:** Interference pattern of a single wave with defined n - **Uncertainty principle:** Practical measurement limit, not fundamental

9.3 Recovery of Einsteinian realism

VED fulfills Einstein's dream: a theory where: 1. An objective reality exists independent of the observer
2. Reality is deterministic 3. The geometry of space-time is fundamental

10. CONCLUSIONS AND PREDICTIONS

10.1 Achievements of the Einstein-VED theory

1. **Derivation of G from first principles:** G emerges as $G = \frac{R_{VED} \cdot R_e \cdot c^3}{2n\hbar}$
2. **Quantization of G:** $G(n) = G_{\text{exp}} \cdot (2/n)$ with experimental verification for $n = 2$
3. **Unification of scales:** Same law $R = n\hbar/(mc)$ applies from proton ($n = 4$, $R \sim 10^{-15}$ m) to atomic electron ($n \approx 137$, $R \sim 10^{-11}$ m)
4. **Determinism and realism:** Elimination of fundamental probability, recovery of objective reality
5. **Resolution of quantum paradoxes:** Entanglement, wave-particle duality, collapse explained geometrically

10.2 Specific and verifiable predictions

1. **Complete spectrum of stable n :** The theory predicts that there exists a discrete set (finite or countable) of n values that correspond to stable particles.
2. **Relation $n_{\text{electron}} = 1/\alpha$:** The coincidence $n_e \approx 137.036 \approx 1/\alpha$ suggests that the fine-structure constant emerges from the geometry of electron confinement.
3. **Null gravity for $n \rightarrow \infty$:** Photons and pure radiation should exhibit negligible gravitational coupling.
4. **Exact atomic geometries:** The theory allows calculating deterministic electron positions in atoms and molecules.

10.3 Philosophical implications and future direction

The Einstein-VED theory proposes a **paradigm shift**: gravity is not a "fundamental force" but an emergent manifestation of the discrete geometry of space-time. The universe is comprehensible because it has **countable structure on a continuous background**.

Future research lines: 1. Complete determination of the spectrum of stable n 2. Derivation of dynamic equations for transitions between n modes 3. Extension of geometry to non-gravitational interactions 4. Formal connection with complete General Relativity 5. Predictions for quantum foundations experiments

FINAL SUMMARY

The **Einstein-VED** theory demonstrates that:

1. **All mass** obeys $R = \frac{n\hbar}{mc}$ with integer n
2. **Gravity** emerges as $G = \frac{R_{VED} \cdot R_s \cdot c^3}{2n\hbar}$
3. **Matter and energy** are extremes of a continuous confinement spectrum
4. **Determinism** is recovered through pure geometry
5. **The universe** has discrete/countable content in a continuous container

Einstein's dream: a deterministic, realistic and geometric theory that unifies the very small with the very large.